

PRACTICAL NO.6

```
In [1]: import numpy as np
```

```
In [2]: from sympy.geometry import *
```

```
In [17]: from sympy import *
```

Q.1)Rotation of a point and shape

```
In [18]: #point
```

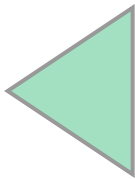
```
In [19]: P=Point(1,2)
P.rotate(np.pi/2)
```

```
Out[19]: Point2D(-2,1)
```

```
In [20]: #Triangle
```

```
In [21]: T=Triangle(Point(0,0),Point(4,0),Point(2,3))
T.rotate(np.pi/2)
```

```
Out[21]:
```



Q.2)Reflect a point and shape

```
In [22]: #Point
```

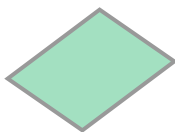
```
In [23]: P.reflect(Line(Point(1,0),Point(0,1)))
```

```
Out[23]: Point2D(-1,0)
```

```
In [24]: #Polygon
```

```
In [25]: P=Polygon(Point(0,0),Point(4,0),Point(5,3),Point(1,3))
P.reflect(Line(Point(1,0),Point(0,3)))
```

```
Out[25]:
```



Q.3)Scaling of a point and shape

```
In [26]: #Point
```

```
In [27]: P=Point(2,3)
P.scale(2,1)
```

```
Out[27]: Point2D(4,3)
```

```
In [28]: #Polygon
```

```
In [29]: poly=Polygon(Point(1,1),Point(4,1),Point(2,3),Point(2,4))
poly.scale(2,2)
```

```
Out[29]:
```



Q.4)Shearing of a point and shape

```
In [30]: #Point
```

```
In [31]: P=Point(3,4)
Sh=Matrix([[1,2,0],[2,1,0],[0,0,1]])
P.transform(Sh)
```

```
Out[31]: Point2D(11,10)
```

```
In [32]: #Polygon
```

```
In [39]: shape=Polygon((0,0),(4,0),(2,3))
shx=1
sheared_shape=Polygon(*[(x+shx*y,y) for x,y in shape.vertices])
print(sheared_shape)
```

```
Triangle(Point2D(0, 0), Point2D(4, 0), Point2D(5, 3))
```

Q.5)Translation of a Point and Shape

```
In [34]: #Point
```

```
In [40]: P=Point(2,3)
P_tran=P.translate(5,4)
print("Translated Point:",P_tran)
```

```
Translated Point: Point2D(7, 7)
```

```
In [41]: shape=Triangle((0,0),(4,0),(2,3))
translated_shape=shape.translate(5,4)
print("Translated Shape:",translated_shape)
```

```
Translated Shape: Triangle(Point2D(5, 4), Point2D(9, 4), Point2D(7, 7))
```

In []: